

source. To achieve this, conventional lights are replaced with Li-Fi-enabled ones. Reliable Li-Fi connectivity requires multiple light sources acting as access points, similar to how Wi-Fi routers are distributed in office settings, according to Deepak.

Using Li-Fi is as simple as plugging a Li-Fi dongle into a USB port, just like a flash drive, to access the internet via light. The focus is currently on laptops and smart devices, with plans to integrate Li-Fi into smartphones in the future. LC Link is primarily targeted at telecom operators and ISPs, while enterprises are the key customers for Li-Fi solutions.

Velmenni, derived from an Icelandic word meaning 'smart system,' has its R&D and registered offices in Delhi.

According to Deepak, Velmenni's technology exclusively uses light for data transmission. Their patent covers several critical components, including the modulation of signals through light and the design of optical systems to extend operational ranges. The patent thoroughly addresses all key aspects of this cutting-edge technology.

Li-Fi relies on a direct line of sight between the light source and the laptop. If the connection is completely obstructed, the signal will be disrupted, leading to a loss of connection. However, Deepak shares that ongoing research is exploring ways to enable Li-Fi networks to function in non-line-of-sight situations by using light reflections from walls and other surfaces within the room.

The startup collaborates with contract manufacturers for product manufacturing while retaining control over quality control (QC) and final assembly at its facility. They also partner with various vendors to produce different components. These parts are sent to the



Li-Fi Access Point Standalone



Li-Fi Dongle



Li-Fi-enabled LED Luminaire

startup, where the final product is assembled, programmed, and then delivered to the end customer. The startup is responsible for selling the finished product and currently employs almost 20 team members.

Building a deep tech company around innovative, unprecedented technology is challenging, particularly when it involves convincing customers to adopt it. In India, there is often a reluctance to embrace new technologies first, unlike in first-world countries where early adoption is common. Initially, there was strong interest from international markets like the US and Europe, which Velmenni leveraged to engage Indian customers. Over time, the mindset in India has shifted, with telecom customers now showing a more positive response. The focus was on R&D for the first five to six years, and after lengthy trials, the commercial product is finally gaining acceptance, overcoming the challenges of time and mindset change, shares Deepak.

While discussing plans, Deepak mentions that they are developing a new solution and product that will revolutionise how internet access works at home, making remote work more seamless. He adds that they are likely to announce it soon and will provide more details at the appropriate time. Deepak also indicates that they plan to launch initially in the international market and then introduce in the Indian market.

The startup has deployed its products in Germany, France, the US,



Deepak Solanki, CEO, Velmenni

and various countries across Asia, including Sri Lanka, India, and Saudi Arabia. Additionally, deployments have been completed in Estonia and several other European countries.

Deepak emphasises that the next two years are vital for their financial growth. Following years of R&D and a pre-revenue phase, the company has achieved 100% year-on-year revenue growth over the past two years. This year, they aim to quintuple revenue and sustain this momentum for continued growth.

The startup asserts that it has developed a robust vendor ecosystem over the past decade, collaborating closely with various partners in the electronics industry. However, they recognise the scaling challenges and are eager to bring on additional partners to support their growth. They are open to forming partnerships within India and internationally and are willing to share revenue opportunities with those who collaborate with them. **EFY**

—Nitisha Dubey